

VNet-to-VNet VPN

using Azure VPN Gateway

Prepared by: Tushar Singh

Date: April 11, 2026

Platform: Microsoft Azure

Objective

To connect two Azure Virtual Networks (VNet-to-VNet) securely using Azure VPN Gateway and verify communication between virtual machines (VMs) using private IP addresses across an encrypted tunnel.

Step 1: Create Resource Group

A dedicated resource group was created in Azure to logically group and manage all lab resources under a single unit.

Resource Group	rg-vnet-vpn-lab
Location	Central India
Subscription	Azure Subscription 1

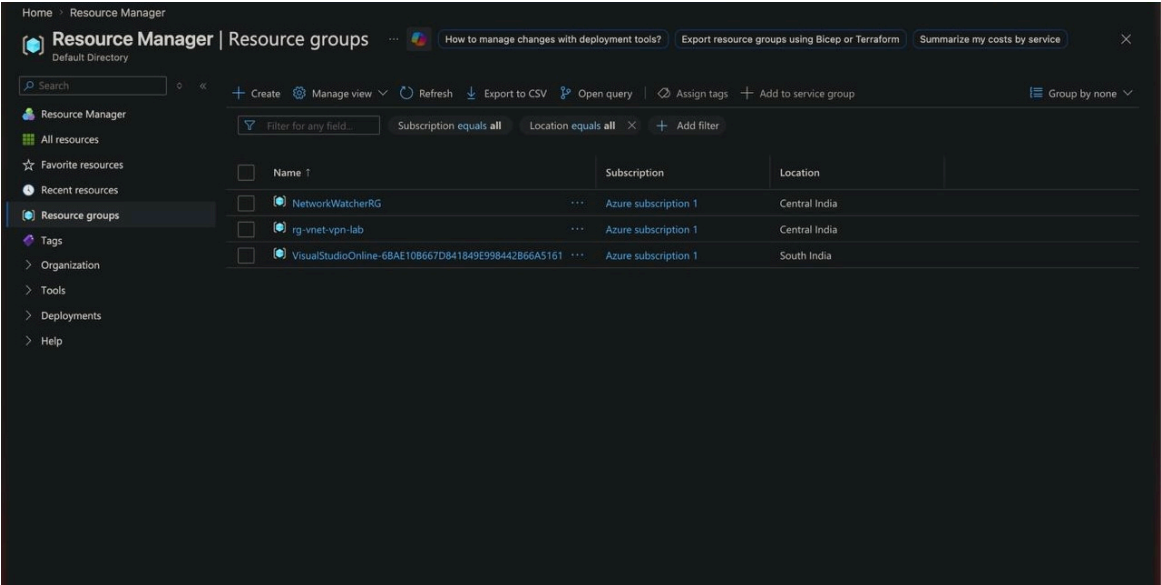


Figure 1: Resource Groups in Azure Portal — rg-vnet-vpn-lab created

Step 2: Create Virtual Networks

Two virtual networks were created with non-overlapping address spaces to support VPN gateway peering.

VNet 1 Name	vnet1
vnet1 Address Space	10.0.0.0/16
VNet 2 Name	VNet-2
VNet-2 Address Space	10.1.0.0/16
Location (both)	Central India

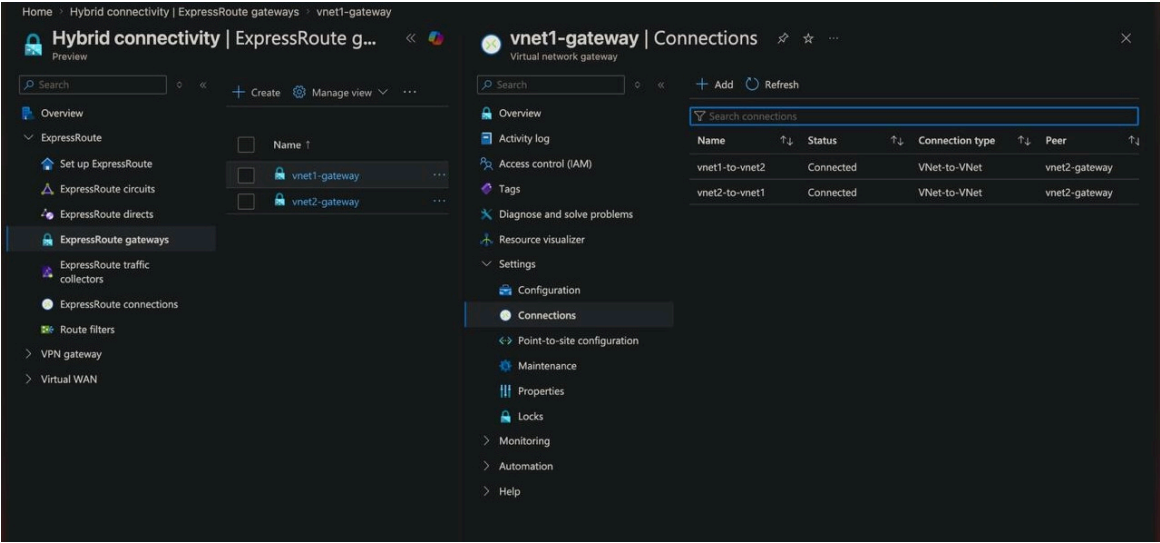


Figure 2: Virtual Networks — vnet1 and VNet-2 listed in Azure Portal

Step 3: Configure Subnets

Each virtual network required two subnets to operate correctly:

- subnet1 — Hosts the virtual machines (VMs) for connectivity testing.
- GatewaySubnet — A dedicated subnet reserved exclusively for the VPN Gateway. Azure requires this subnet; it cannot host any other resources. Minimum prefix: /27.

Step 4: Create VPN Gateways

A VPN Gateway was provisioned inside the GatewaySubnet of each virtual network. These gateways handle authentication, key exchange, and encrypted tunnel establishment between the VNets.

4a. VPN Gateway List

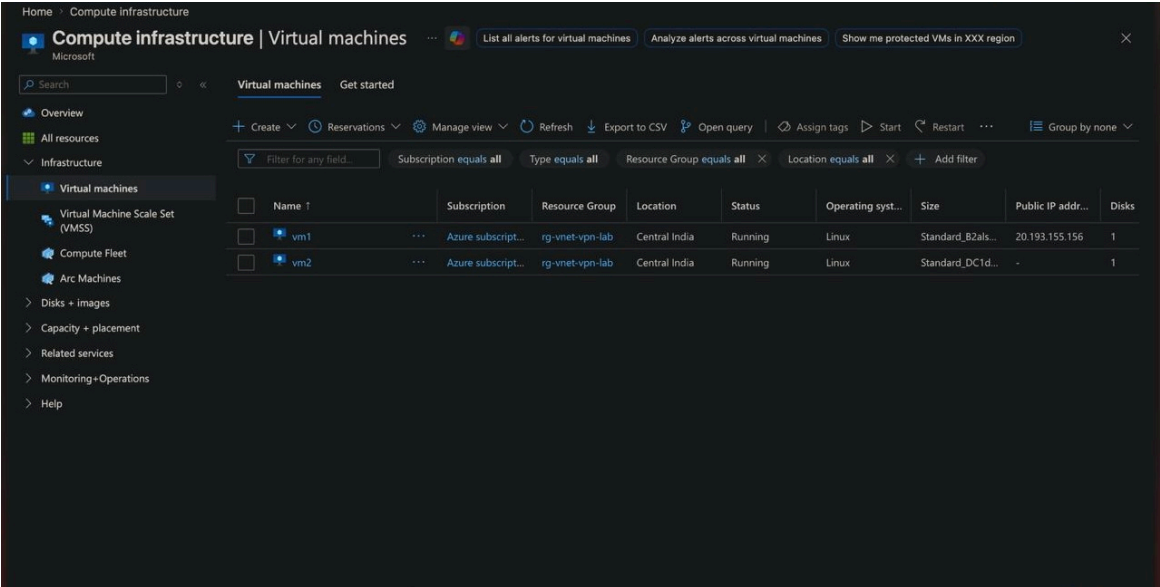


Figure 3: ExpressRoute Gateways — vnet1-gateway and vnet2-gateway listed

4b. vnet1-gateway — Configuration Details

Gateway Name	vnet1-gateway
Virtual Network	vnet1
Gateway Type	VPN
VPN Type	Route-based
SKU	VpnGw1AZ
Public IP Address	20.204.221.108 (pip-vnet1-gw)
Location	Central India

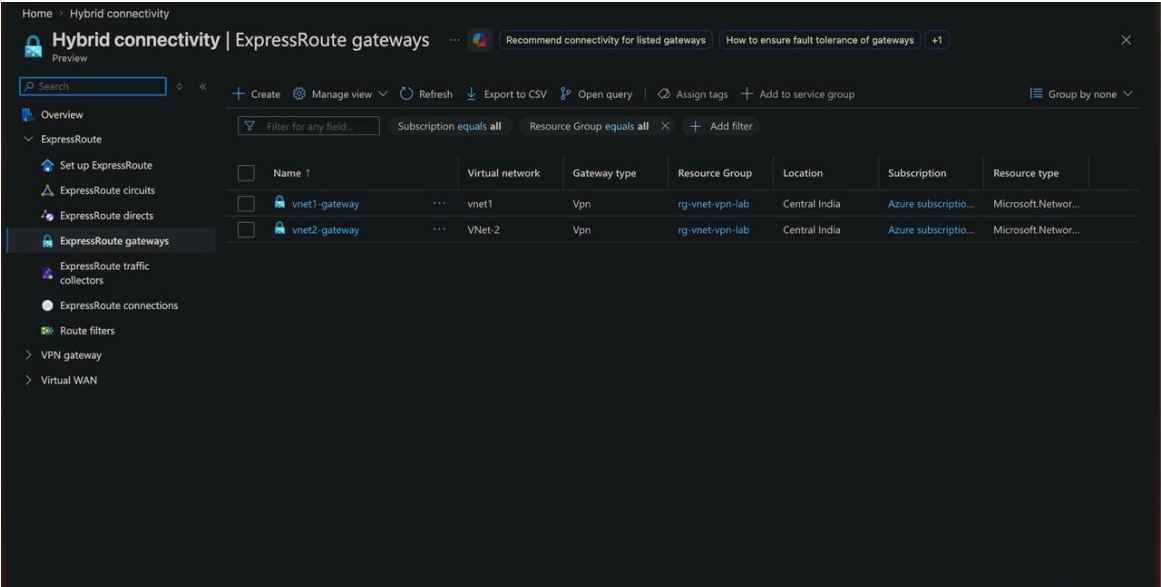


Figure 4: vnet1-gateway — Overview and configuration in Azure Portal

4c. vnet2-gateway — Configuration Details

Gateway Name	vnet2-gateway
Virtual Network	VNet-2
Gateway Type	VPN
VPN Type	Route-based
SKU	VpnGw1AZ
Public IP Address	20.219.211.251 (pip-vnet2-gw)
Location	Central India

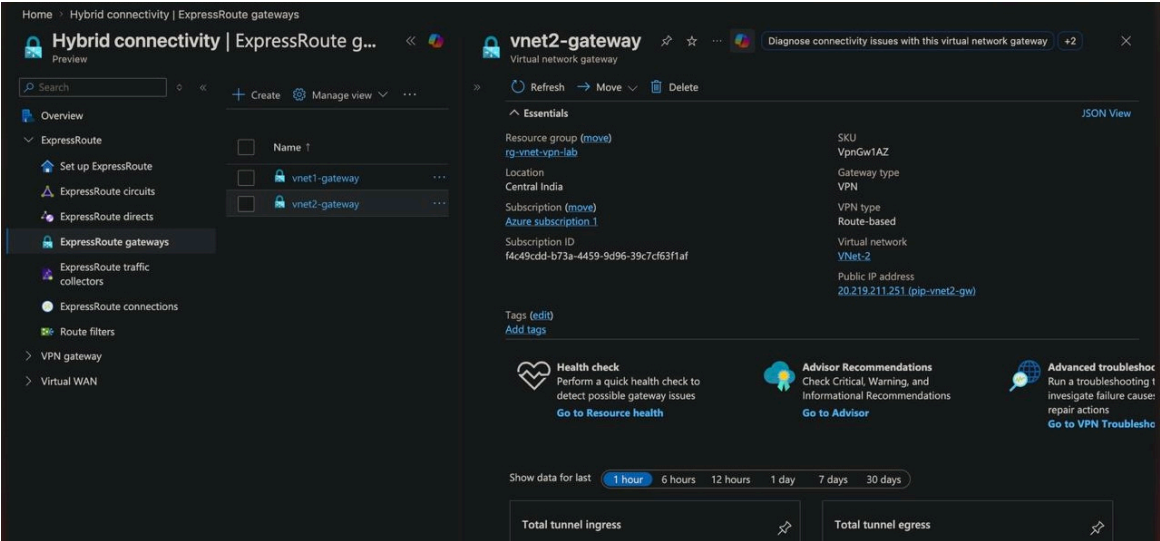


Figure 5: vnet2-gateway — Overview and configuration in Azure Portal

Step 5: Create VNet-to-VNet Connections

Two VNet-to-VNet connections were configured using a shared pre-shared key to establish the encrypted bidirectional IPsec/IKE tunnel.

- Connection 1: vnet1-to-vnet2 → from vnet1-gateway to vnet2-gateway
- Connection 2: vnet2-to-vnet1 → from vnet2-gateway to vnet1-gateway
- Connection Type: VNet-to-VNet

✓ **Status: Connected** — Both connections successfully established and showing "Connected" in Azure Portal

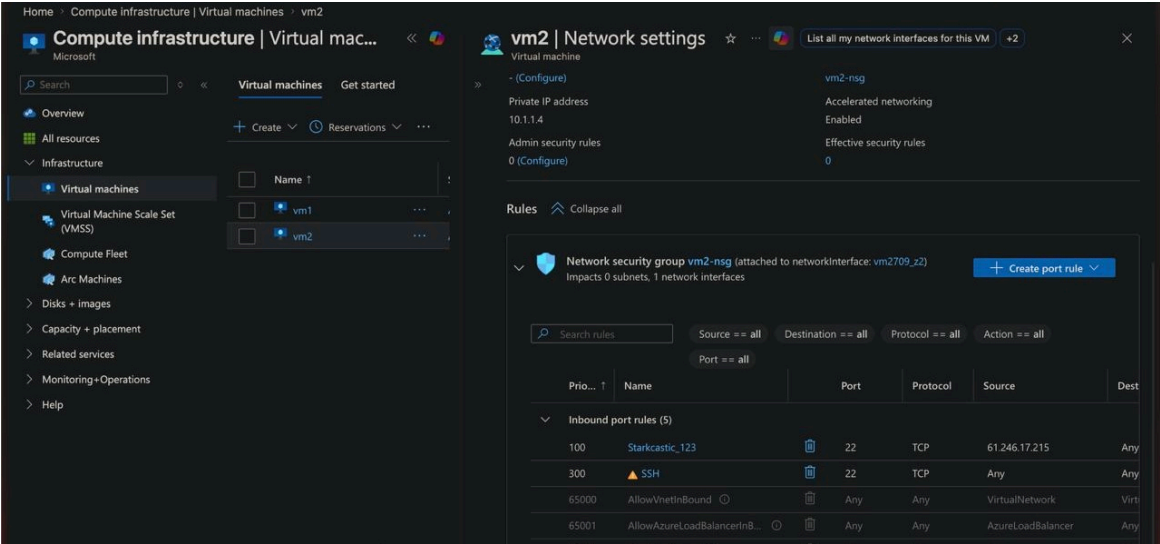


Figure 6: vnet1-gateway Connections — Both showing "Connected" status

Step 6: Create Virtual Machines

One Linux virtual machine was deployed in each virtual network subnet to enable cross-VNet connectivity testing.

VM 1 Name	vm1
vm1 VNet / Subnet	vnet1 / subnet1
vm1 Private IP	10.0.1.4
VM 2 Name	vm2
vm2 VNet / Subnet	VNet-2 / subnet1
vm2 Private IP	10.1.1.4
Operating System	Ubuntu 24.04.4 LTS
Status (both)	Running

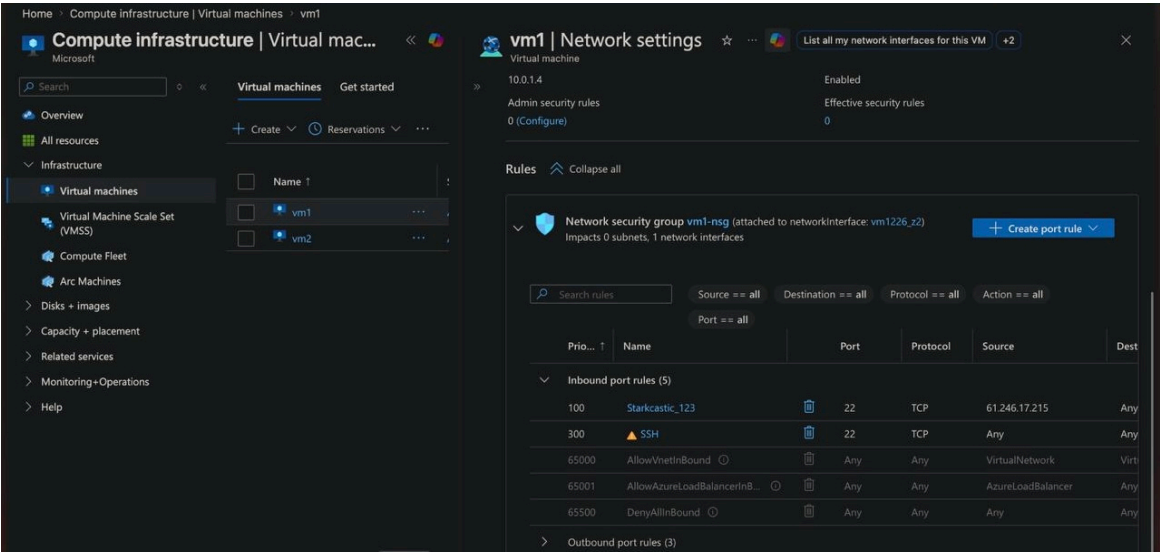


Figure 7: Virtual Machines — vm1 and vm2 running in Central India

6a. vm1 — Network Security Group Rules

The NSG attached to vm1 (vm1-nsg) was configured with an inbound rule allowing SSH on port 22, plus the default Azure rules for VNet and Load Balancer traffic.

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Sat Apr 11 06:09:12 UTC 2026

System load: 0.0          Processes: 131
Usage of /: 5.7% of 28.02GB Users logged in: 0
Memory usage: 7%          IPv4 address for eth0: 10.0.1.4
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm1:~$ ls
azureuser@vm1:~$ pwd
/home/azureuser
azureuser@vm1:~$ ping -c 4 10.1.1.4
PING 10.1.1.4 (10.1.1.4): 56(84) bytes of data:
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=2.32 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=35.1 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=1.61 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=1.70 ms

--- 10.1.1.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.613/10.184/35.099/14.387 ms
azureuser@vm1:~$ exit
logout
```

Figure 8: vm1 NSG (vm1-nsg) — Inbound port rules: SSH (port 22) allowed

6b. vm2 — Network Security Group Rules

The NSG attached to vm2 (vm2-nsg) mirrors the same ruleset, permitting inbound SSH traffic on port 22 across the VPN tunnel.

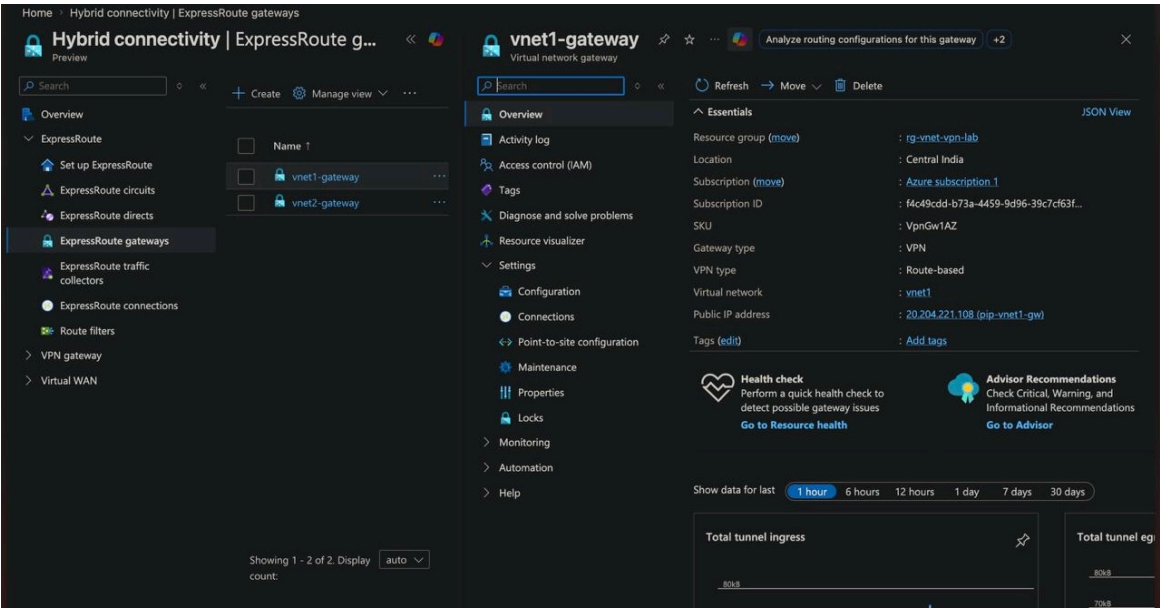


Figure 9: vm2 NSG (vm2-nsg) — Inbound port rules: SSH (port 22) allowed

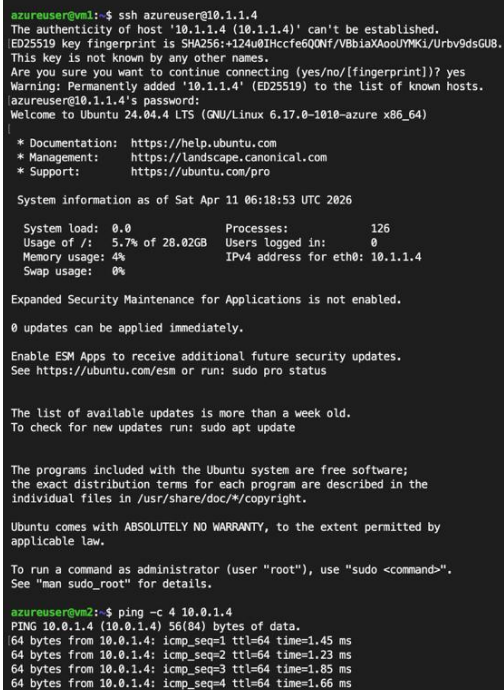
Step 7: Test Connectivity via Ping

ICMP ping tests were performed between the two virtual machines using their private IP addresses to confirm bidirectional traffic through the VPN tunnel.

7a. Ping from VM1 → VM2 (10.0.1.4 → 10.1.1.4)

Executed on vm1:

```
azureuser@vm1:~$ ping -c 4 10.1.1.4
```



```
azureuser@vm1:~$ ssh azureuser@10.1.1.4
The authenticity of host '10.1.1.4 (10.1.1.4)' can't be established.
ED25519 key fingerprint is SHA256:+124u0IHccfe6QONf/VBb1aXaouYMK1/Urbv9dsGU8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.1.1.4' (ED25519) to the list of known hosts.
azureuser@10.1.1.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 11 06:18:53 UTC 2026

System load:  0.0               Processes:    126
Usage of /:   5.7% of 28.02GB   Users logged in:  0
Memory usage: 4%               IPv4 address for eth0: 10.1.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm2:~$ ping -c 4 10.0.1.4
PING 10.0.1.4 (10.0.1.4) 56(84) bytes of data:
64 bytes from 10.0.1.4: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.1.4: icmp_seq=2 ttl=64 time=1.23 ms
64 bytes from 10.0.1.4: icmp_seq=3 ttl=64 time=1.85 ms
64 bytes from 10.0.1.4: icmp_seq=4 ttl=64 time=1.66 ms
```

Figure 10: vm1 terminal — Ping to 10.1.1.4 (vm2): 4/4 packets received, 0% packet loss

✓ **Result: 4 packets transmitted, 4 received, 0% packet loss**

7b. Ping from VM2 → VM1 (10.1.1.4 → 10.0.1.4)

Executed on vm2:

```
azureuser@vm2:~$ ping -c 4 10.0.1.4
```

```
ED25519 key fingerprint is SHA256:+124u0IHccfe6QONf/VBbiaXaouYMKL/Urbv9dsGU8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.1.1.4' (ED25519) to the list of known hosts.
azureuser@10.1.1.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr 11 06:18:53 UTC 2026

System load:  0.0          Processes:    126
Usage of /:   5.7% of 28.02GB Users logged in:   0
Memory usage: 4%          IPv4 address for eth0: 10.1.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm2:~$ ping -c 4 10.0.1.4
PING 10.0.1.4 (10.0.1.4) 56(84) bytes of data:
64 bytes from 10.0.1.4: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.1.4: icmp_seq=2 ttl=64 time=1.23 ms
64 bytes from 10.0.1.4: icmp_seq=3 ttl=64 time=1.85 ms
64 bytes from 10.0.1.4: icmp_seq=4 ttl=64 time=1.66 ms

--- 10.0.1.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.234/1.549/1.851/0.230 ms
```

Figure 11: vm2 terminal — Ping to 10.0.1.4 (vm1): 4/4 packets received, 0% packet loss

✓ **Result: 4 packets transmitted, 4 received, 0% packet loss**

Step 8: SSH Cross-VNet Connection

An SSH connection was initiated from vm1 to vm2 using vm2's private IP address (10.1.1.4), confirming full Layer-4 TCP connectivity through the encrypted VPN tunnel.

```
azureuser@vm1:~$ ssh azureuser@10.1.1.4
```

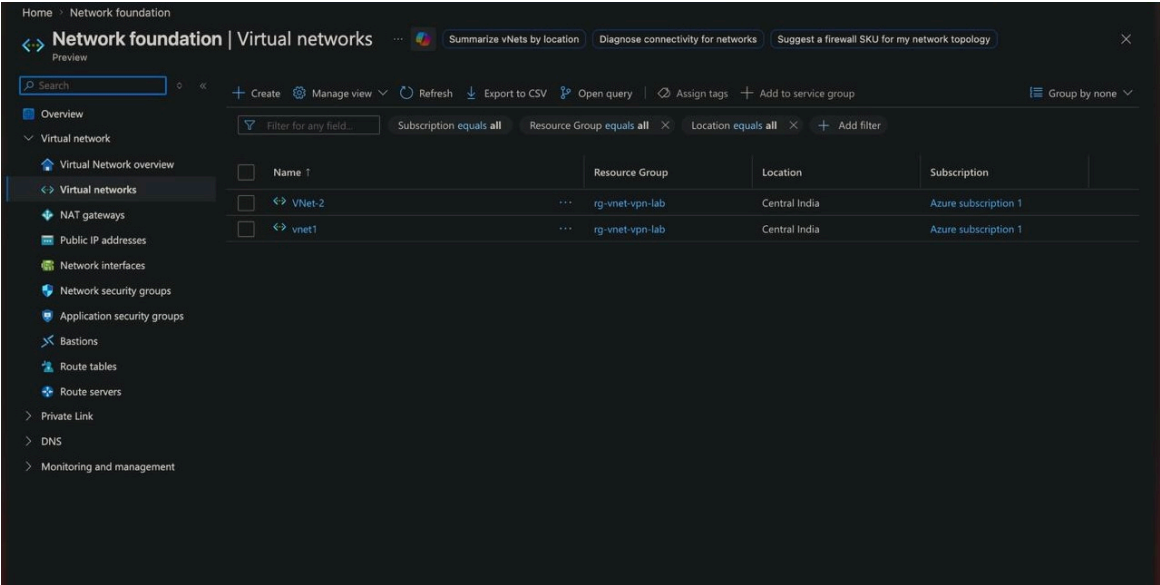


Figure 12: vm1 terminal — SSH successfully connected to vm2 at 10.1.1.4

✓ SSH connection established from vm1 to vm2 across the VPN tunnel using private IP

Result & Conclusion

The VNet-to-VNet VPN lab was completed successfully. All objectives were achieved:

- Two Azure Virtual Networks (vnet1 and VNet-2) were created with non-overlapping address spaces.
- A VPN Gateway was provisioned in the GatewaySubnet of each VNet.
- VNet-to-VNet connections were configured with a shared key; both showed "Connected" status.
- Ping tests from vm1 to vm2 and vm2 to vm1 returned 0% packet loss over private IPs.
- SSH connectivity from vm1 → vm2 was verified across the encrypted tunnel using private IP 10.1.1.4.

The secure VPN tunnel ensures all inter-VNet traffic is encrypted end-to-end via IPsec/IKE, enabling private and secure communication without exposing resources to the public internet.